

1.6 Series

Pure Mathematics 1 · 2026 examination · Concept guide and original exam-style practice

Understand the concept

Arithmetic sequences have $u_n = a + (n-1)d$ and $S_n = n[2a + (n-1)d]/2$. A geometric sequence has $u_n = ar^{n-1}$, $S_n = a(1-r^n)/(1-r)$ if $r \neq 1$, and infinite sum $a/(1-r)$ only for $\text{abs}(r) < 1$.

Worked example: follow the reasoning

For 3, 6, 12, ..., $a=3, r=2$ and fifth term 48; no finite infinite sum because $\text{abs}(r) > 1$.

Structured practice

Answer in the space provided. Show working and units where relevant.

1 Find first six terms' sum for geometric series 2, 6, 18, ...

[3]

Final answer: _____

Answers and explanation

Question 1 [3 marks]

$a=2$, $r=3$ [1]; $S_6=2(3^6-1)/(3-1)$ [1] = 728 [1].

Self-assessment

Compare each credited step with your own answer. Fix the first reasoning or calculation error; then retry without looking at the solution.

Syllabus: Mathematics 9709, 2026–2027. <https://www.cambridgeinternational.org/Images/697427-2026-2027-syllabus.pdf>

Original FagusED teaching material. Question layout and mark presentation informed by the May/June 2026 papers supplied by the user. The practice question is newly written and is not a Cambridge past-paper question.